

PHYS 230 - Electricity and Magnetism

Lecture 20

Instructor: Saeed Rastgoo



**UNIVERSITY
OF ALBERTA**

Outline

4 Capacitance and Dielectrics

- Dielectrics (Continued)

All figures in these slides are from one of the following sources, unless cited otherwise:

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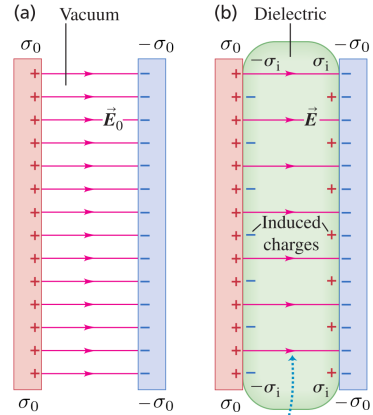
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Electric Field in a Capacitor With a Dielectric

Replacing air by a dielectric with dielectric constant K , but maintaining the same Q in a capacitor

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.

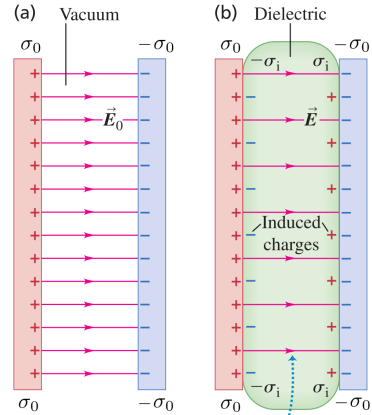


For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

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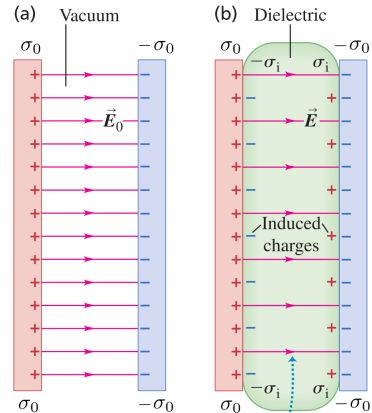
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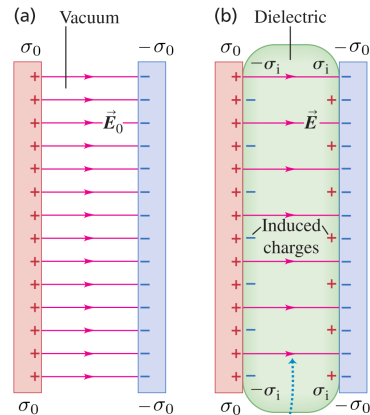
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But in a parallel plate capacitor with plate separation d we have
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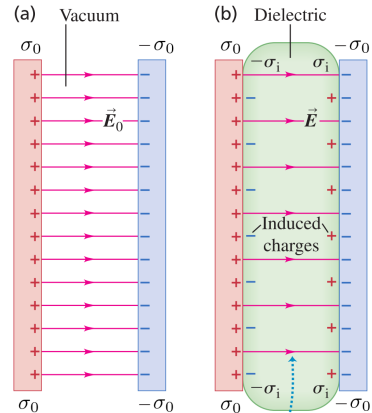
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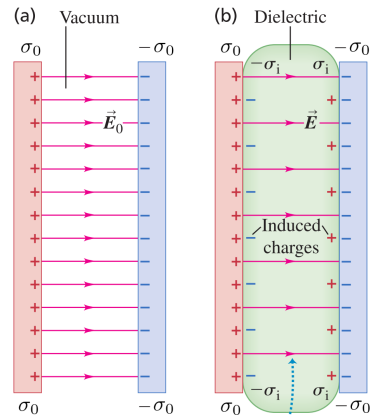
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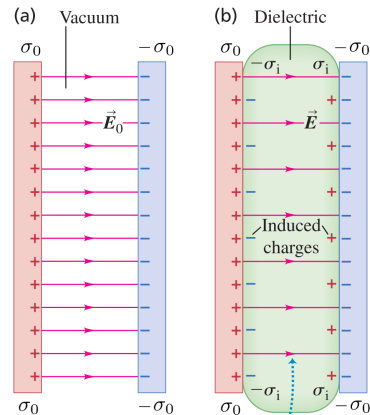
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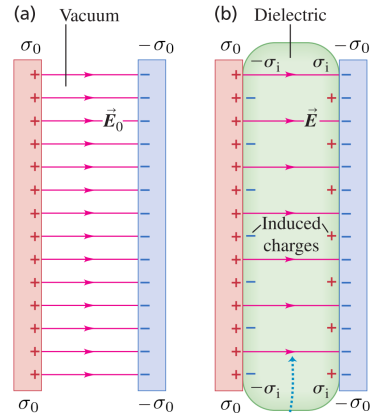
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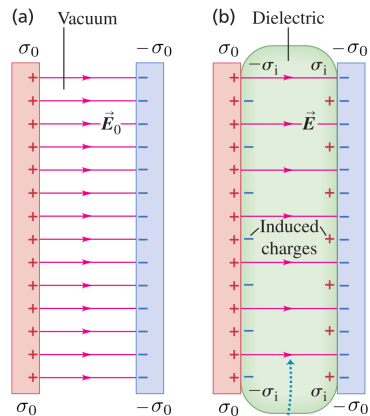
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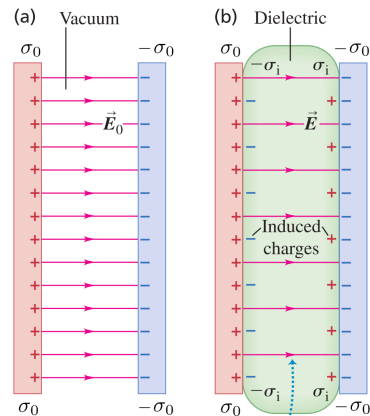
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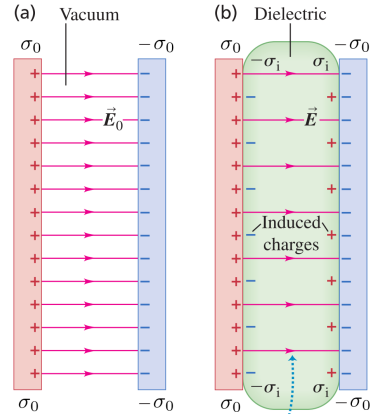
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On the other hand we know that:

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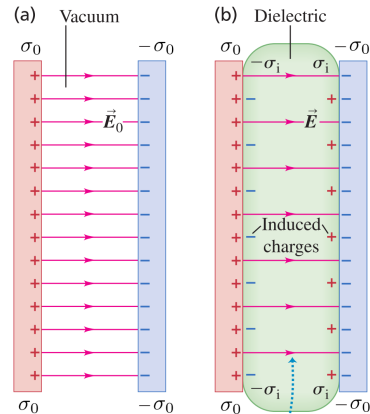
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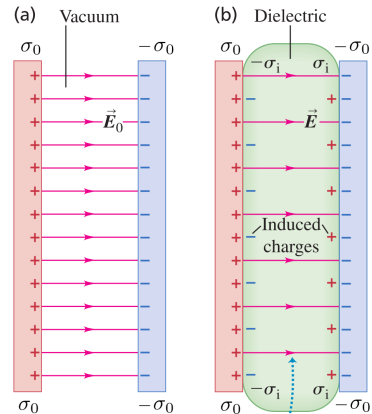
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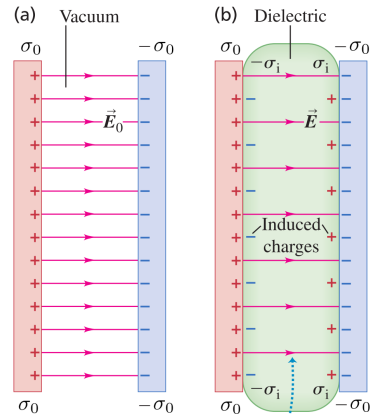
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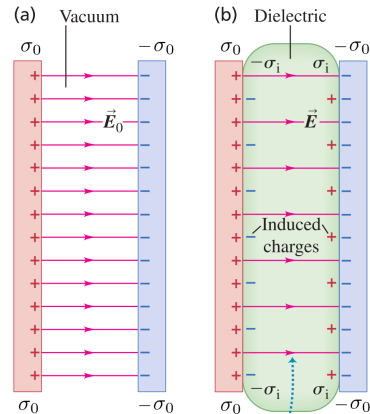
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$$|\sigma| = |\sigma_0 - \sigma_i| < |\sigma_0|$$

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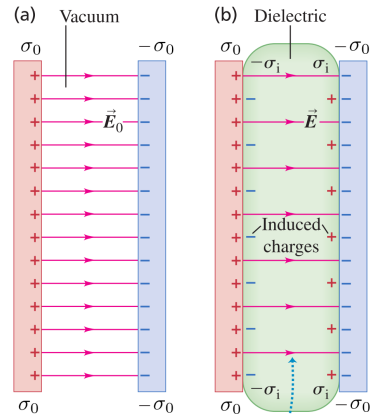
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In other words

$$\begin{cases} \text{on the positive side: } \sigma_0 - \sigma_i \\ \text{on the negative side: } -\sigma_0 + \sigma_i \end{cases}$$

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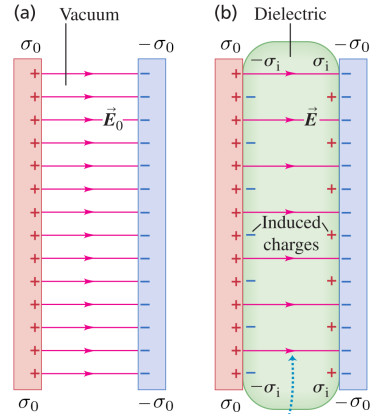


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- The dielectric was originally electrically neutral and is still neutral

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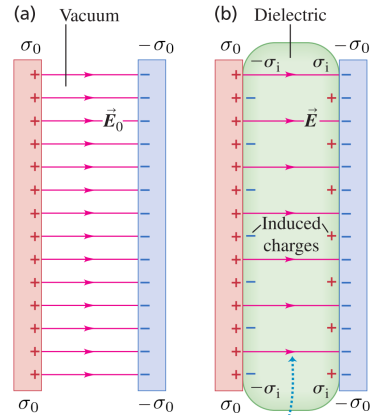


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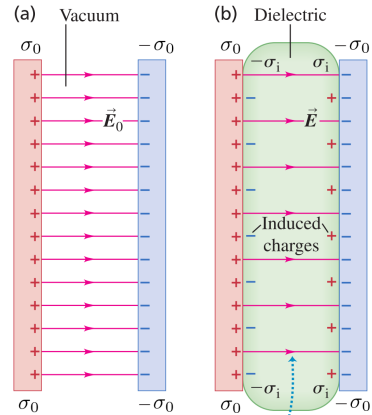


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Electric Field in a Capacitor With a Dielectric

- The dielectric was originally electrically neutral and is still neutral
- The induced surface charges arise as a result of redistribution of positive and negative charges within the dielectric material
- This phenomenon called **polarization**

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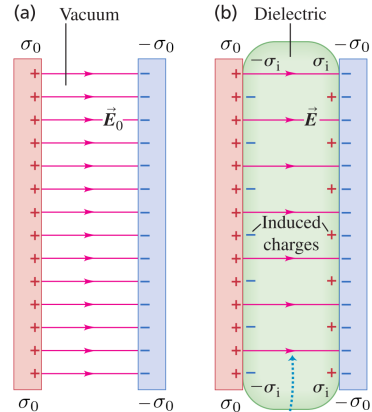
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From the previous slides, on the positive plate

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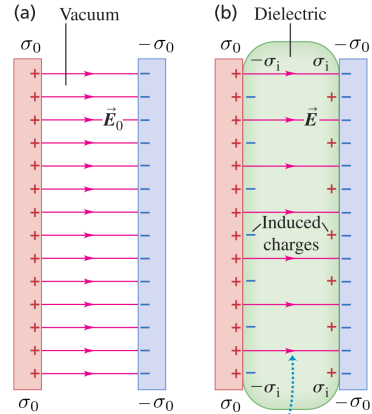
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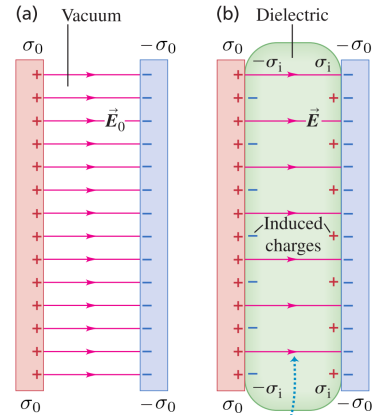
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Combing these we get

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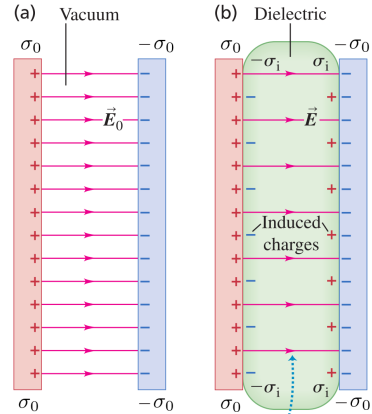
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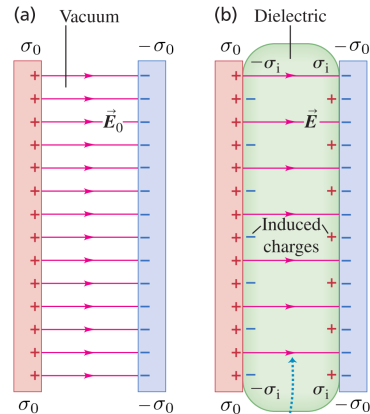
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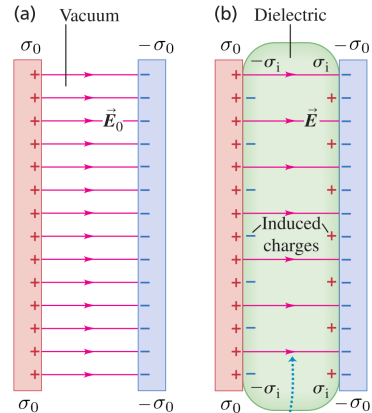
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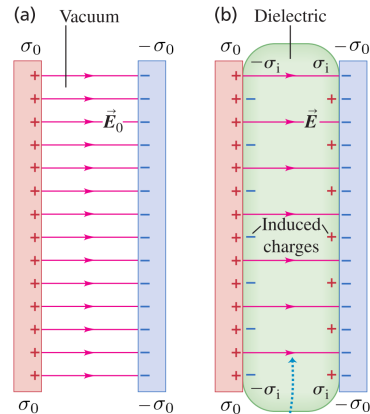
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$$\frac{1}{K} \approx 0$$

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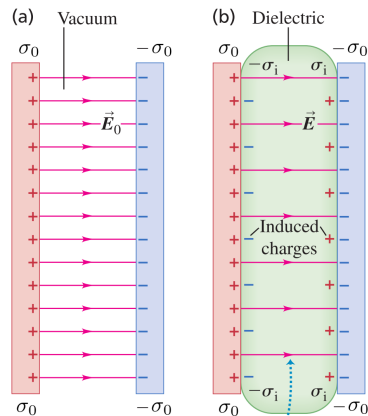
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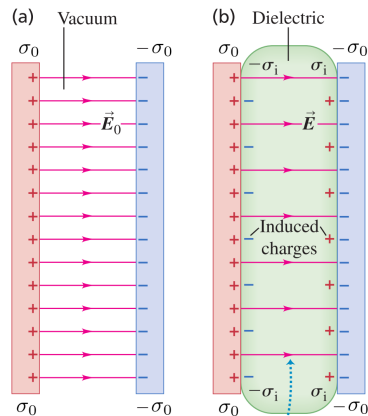
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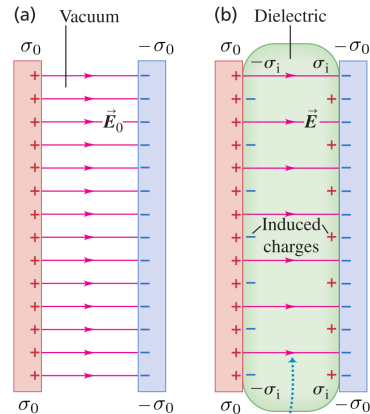
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If K is very large, i.e., $K \gg 1$, then

$$\frac{1}{K} \approx 0 \Rightarrow \sigma_i \approx \sigma_0 \Rightarrow \sigma = \sigma_0 - \sigma_i \approx 0$$

Then $|\mathbf{E}| = \frac{\sigma_0 - \sigma_i}{\epsilon_0}$ and $V = |\mathbf{E}| d$ are much smaller than the vacuum case

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.



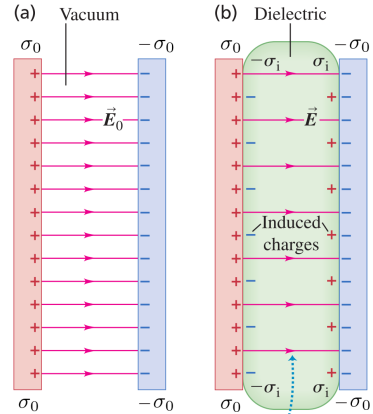
For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

Permittivity of a Dielectric

If we combine

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(a) vacuum between the plates and
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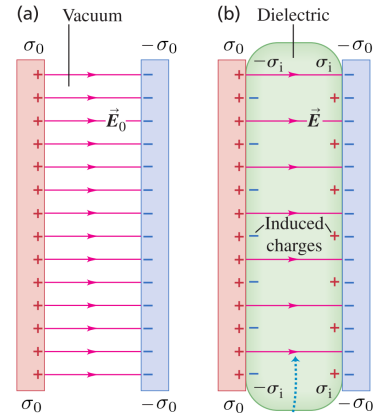
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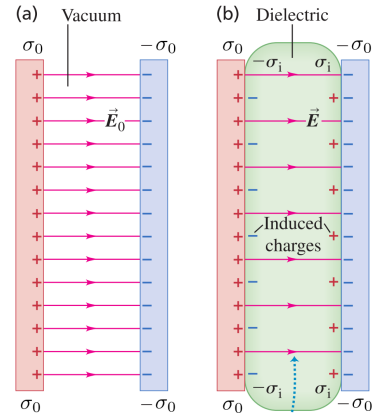
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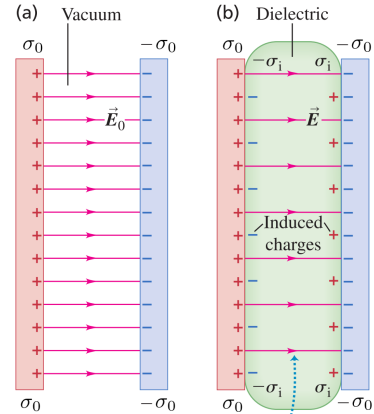
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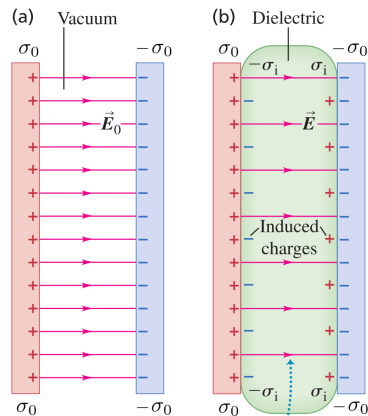
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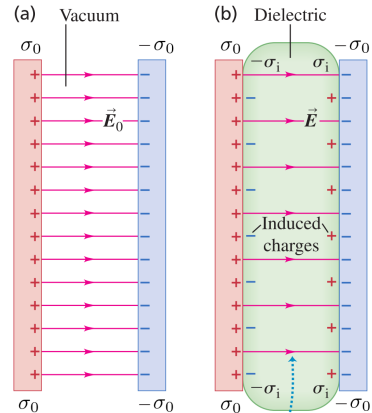
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and hence

$$|\mathbf{E}| = \frac{\sigma_0}{K\epsilon_0}$$

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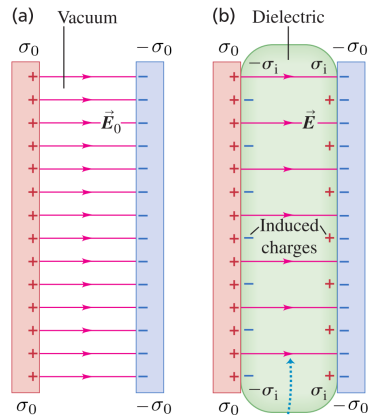
and hence

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We can then define the **permittivity of the dielectric, ϵ** , as

$$\epsilon = K\epsilon_0$$

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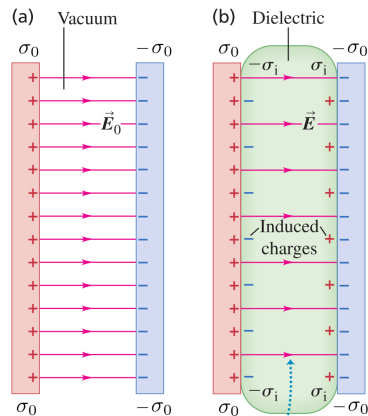
We can then define the **permittivity of the dielectric, ϵ** , as

$$\boxed{\epsilon = K\epsilon_0}$$

which means that we can also write the above $|\mathbf{E}|$ as

$$\boxed{|\mathbf{E}| = \frac{\sigma_0}{\epsilon}}$$

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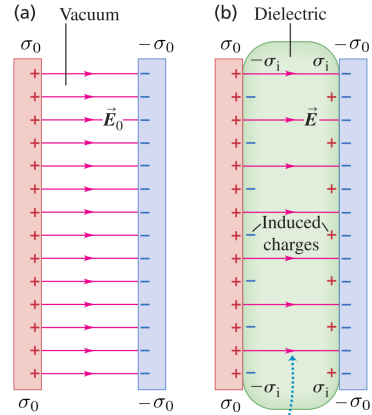
For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

Capacitance With a Dielectric

Since $\frac{V_0}{V} = K$ the new capacitance with a dielectric with constant K with the same charge Q as in the vacuum case is

$$C = \frac{Q}{V}$$

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.



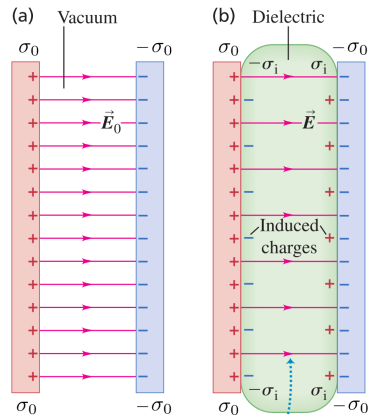
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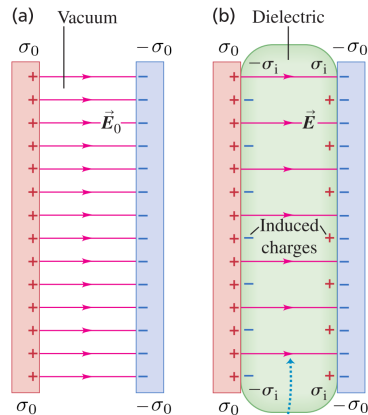
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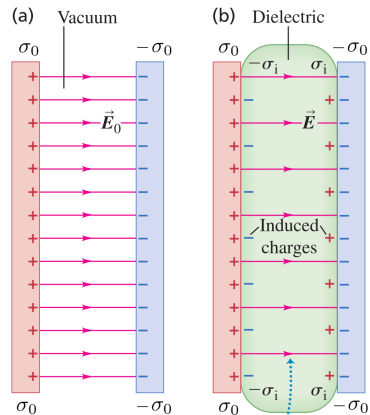
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For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

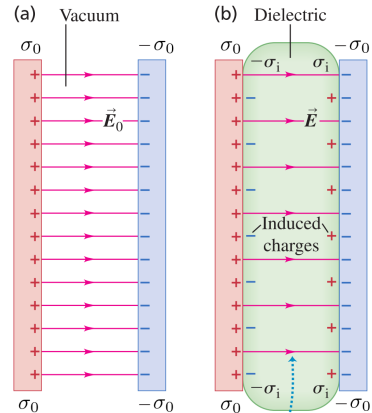
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But since $C_0 = \epsilon_0 \frac{A}{d}$ we get

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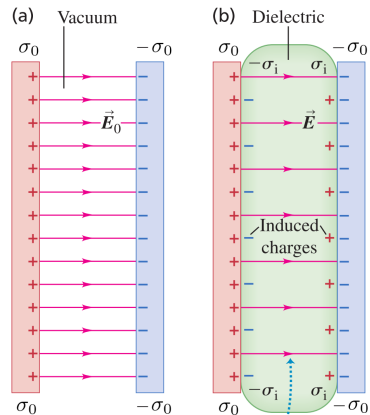
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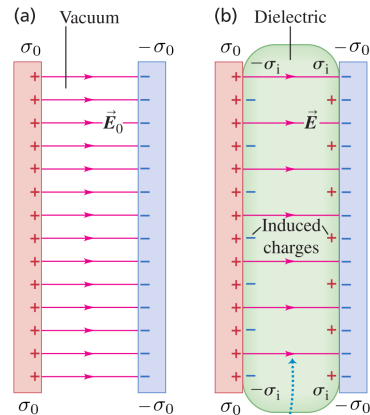
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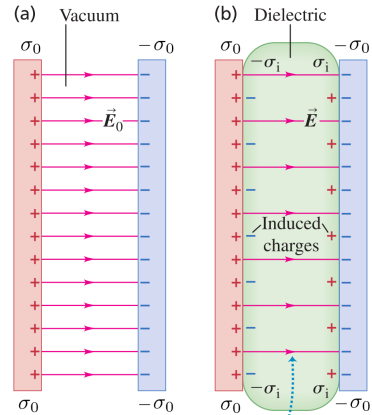
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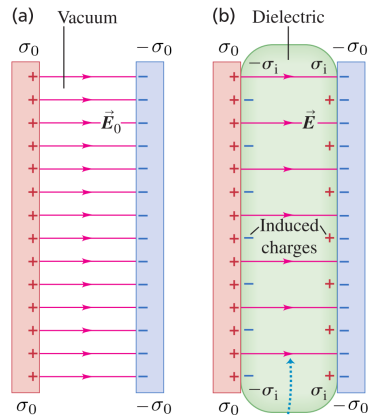
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Also since $E = \frac{V}{d}$ for a parallel plate capacitor, the energy density the electric field within its dielectric is

$$u = \frac{U}{V}$$

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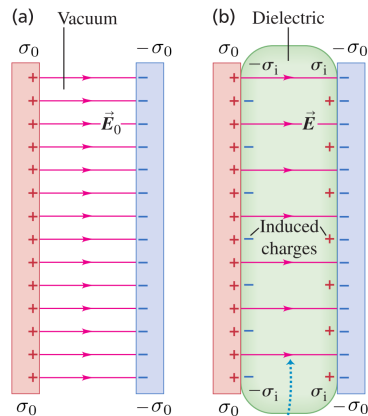
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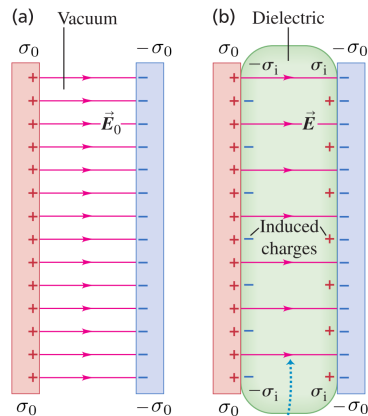
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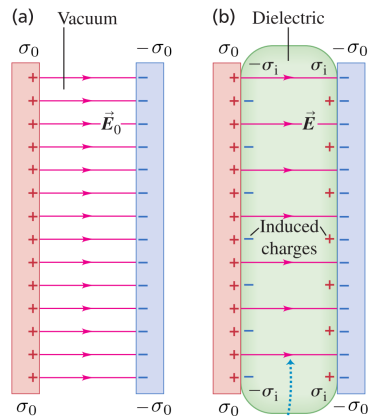
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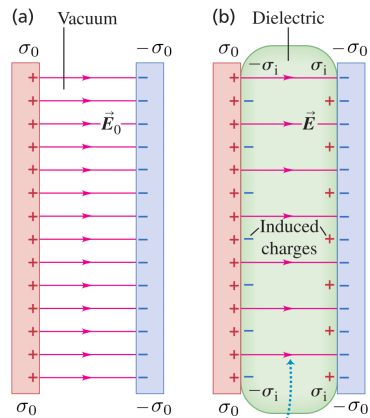
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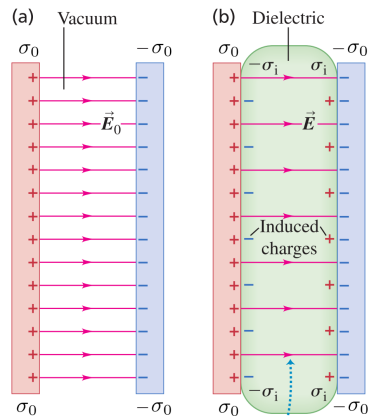
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Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.



For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

Capacitance With a Dielectric

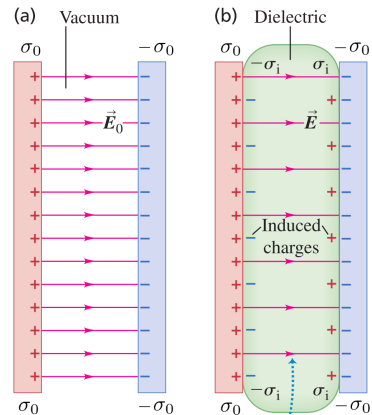
So in summary

$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d}$$

and

$$u = \frac{1}{2}K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2$$

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For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

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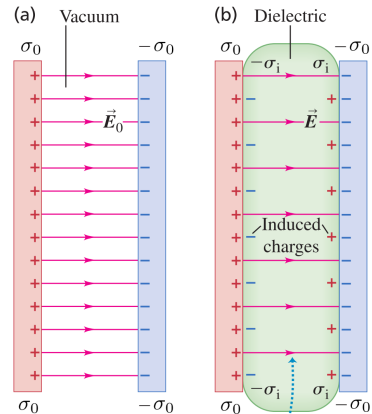
$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d}$$

and

$$u = \frac{1}{2}K\epsilon_0 E^2 = \frac{1}{2}\epsilon E^2$$

And for the **vacuum**, just set $K = 1$.

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.



For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

Capacitance With a Dielectric

So in summary

$$C = KC_0 = K\epsilon_0 \frac{A}{d} = \epsilon \frac{A}{d}$$

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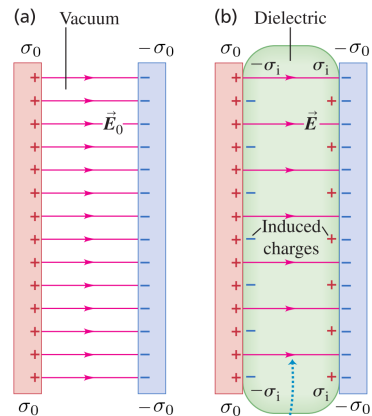
And for the **vacuum**, just set $K = 1$.

Since in vacuum

$$\epsilon = \underbrace{K}_1 \epsilon_0 = \epsilon_0$$

the constant ϵ_0 is also called “permittivity of free space” or the “permittivity of vacuum.”

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.



For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

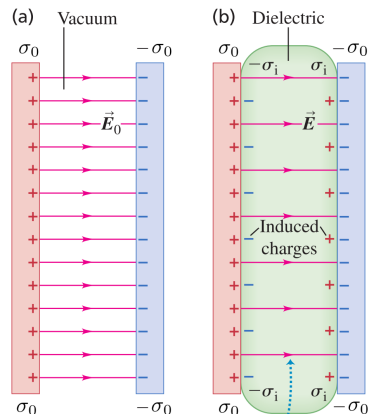
Exercise I

<https://epoll.srv.ualberta.ca> – Code: MBD

Suppose the parallel plates in the figure each have an area of $2 \times 10^{-1} \text{ m}^2$ and are $1 \times 10^{-2} \text{ m}$ apart. We connect the capacitor to a power supply, charge it to a potential difference $V_0 = 3.00 \text{ kV}$, and disconnect the power supply. We then insert a sheet of insulating plastic material between the plates, completely filling the space between them. We find that the potential difference decreases to 1.00 kV while the charge on each capacitor plate remains constant. Find

- (a) the original capacitance C_0
- (b) the magnitude of charge Q_0 on each plate
- (c) the dielectric constant K of the dielectric
- (d) the capacitance C after the dielectric is inserted
- (e) the permittivity ϵ of the dielectric
- (f) the magnitude of the induced charge Q_i on each face of the dielectric
- (g) the original electric field E_0 between the plates
- (h) the magnitude of the electric field E after the dielectric is inserted.

Figure 24.15 Electric field lines with (a) vacuum between the plates and (b) dielectric between the plates.

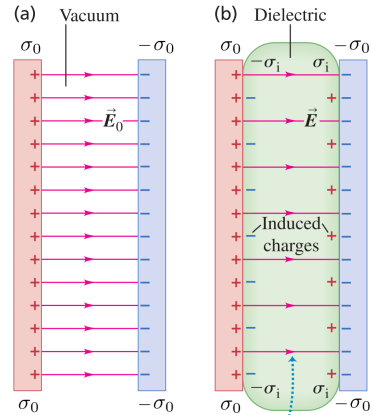


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Exercise I

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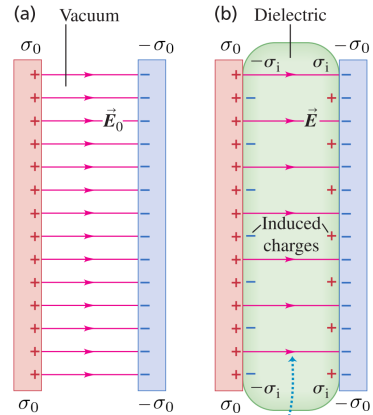


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Exercise I

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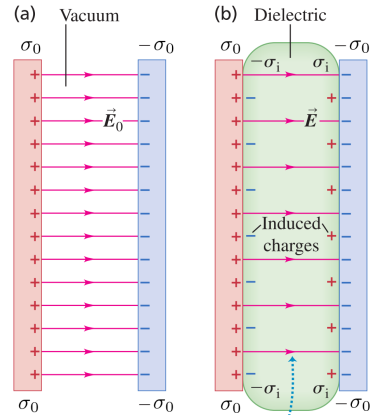


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Exercise I

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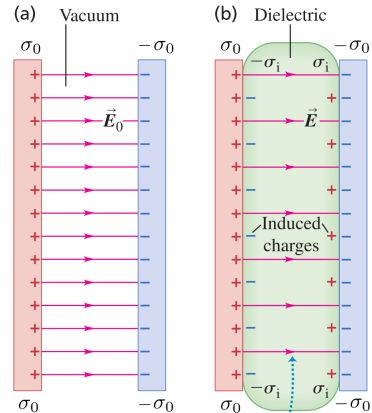
For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

Exercise 2

<https://epoll.srv.ualberta.ca> – Code: MBD

Find the energy and the energy density stored in the electric field of the capacitor in previous example, both before and after the dielectric sheet is inserted.

Figure 24.15 Electric field lines with
(a) vacuum between the plates and
(b) dielectric between the plates.

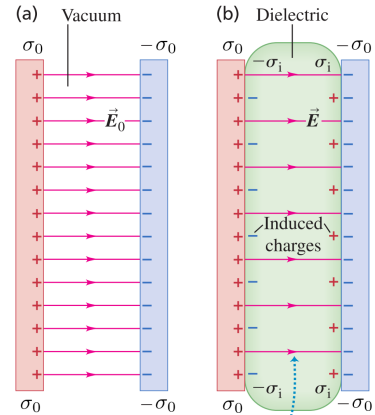


For a given charge density σ , the induced charges on the dielectric's surfaces reduce the electric field between the plates.

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